

Propionate. Anti-obesity and satiety enhancing factor?

Propionate is produced along with acetate and butyrate as a result of fermentative activity of gut microflora on dietary fiber. It has long been known to exhibit hypophagic effects in ruminants, however, its potential physiological roles in non-ruminants as well as humans remained unnoticed over the years. In view of various studies pointing towards the hypophagic as well as hypocholesterolemic effects of propionate in humans, it may act as an important factor in amelioration of obesity, a lifestyle disease arising due to energy imbalance and growing at a startling rate globally. Short chain fatty acids have recently been ascribed as ligands to G-protein coupled receptors (GPRs) 41 and 43. Thus, propionate along with acetate may also be involved in the regulation of adipogenesis and adipokine release mediated via GPRs. The present review summarizes the evidence which collectively raise the possibility of propionate as a dietary factor to depress appetite and combat the obesity epidemic. Copyright © 2011 Elsevier Ltd. All rights reserved.